$x + 1 \times (+6) + 4 \times (-2) = 0$

$x + (+6) + (-8) = 0$

$x = +2$ Therefore, ON of Fe in $FeSO_4$ is +2

Problems:

1. Find the oxidation number of Mn in $KMnO_4$
2. Find the oxidation number of Cr in $Na_2Cr_2O_7$
3. Find the oxidation number of Cu in $CuSO_4$
4. Find the oxidation number of Fe in FeO

Points to Remember

- The tendency of atoms to have eight electrons in the valence shell is known as the 'Octet rule' or the 'Rule of eight'.
- The Lewis dot structure or electron dot symbol for an atom consists of the symbol of the element surrounded by dots representing the electrons of the valence shell of the atom.
- There are different types of chemical bonding possible between atoms which make the molecules. Depending on the type of bond they show different characteristics or properties.
- An ionic bond is formed by the electrostatic attraction between positive and negative ions. It is also called as Electrochemical bond.
- The covalent bond is formed because of the sharing of electrons which become common to both the atoms. It is also called as Atomic bond.
- In some compounds the formation of a covalent bond between two atoms takes place by the sharing of two electrons, both of which comes from only one of the combining atoms. This bond is called Coordinate covalent bond or Dative bond.
- Substances which have the ability to oxidise other substances are called Oxidising agents. These are also called as electron acceptors because they remove electrons form other substances.
- Substances which have the ability to reduce other substances are called Reducing agents. These are also called as electron donors because they donate electrons to other substances.
- Oxidation number also called Oxidation State.

A-Z GLOSSARY

Chemical bond	Force of attraction between the two atoms that binds them together as a unit.
Coordinate covalent bond	Bond formed between atoms by mutual sharing of electrons which are supplied by one atom.
Covalent bond	Bond formed between atoms by the mutual sharing of electrons.
Ionic / Electrovalent bond	Bond formed between cation and anion because of the transfer of electrons from one atom to other atom.
Octet rule or Rule of eight	The tendency of atoms to have eight electrons in the valence shell.
Oxidation	Chemical reaction which involves in the addition of oxygen or removal of hydrogen or loss of electrons.
Oxidation number	The formal charge which an atom has when electrons are counted.
Oxidising agents	Substances which have the ability to oxidise other substances.
Redox reaction	Oxidation and reduction occurs in the same reaction simultaneously.
Reducing agents	Substances which have the ability to reduce other substances.
Reduction	Chemical reaction which involves in the addition of hydrogen or removal of oxygen or gain of electrons.

IX_SCI_EM_Unit-13_CHE.indd 158 12/7/2022 6:14:46 PM